

8(a)	234, 92 for the uranium nucleus	B1
	4, 2 for the alpha particle	B1
8(b)(i)	$N_0 = 0.874 / (238 \times 1.66 \times 10^{-27})$ $= 2.21 \times 10^{24}$	A1
8(b)(ii)	$A = \lambda N$	C1
	$= \frac{\ln 2}{87.7 \times 365 \times 24 \times 3600} \times 2.21 \times 10^{24}$ $= 5.54 \times 10^{14} \text{ Bq}$	A1
8(b)(iii)	power = $5.54 \times 10^{14} \times 5.59 \times 10^{-6} \times 1.60 \times 10^{-19}$	C1
	= 496 W	A1
8(b)(iv)	$65.3 = 100e^{-\frac{\ln 2}{87.7}t}$	C1
	$\ln 0.653 = -(\ln 2 / 87.7) t$	A1
	$t = 53.9 \text{ years}$	
8(c)	advantage: less mass so less energy needed to launch probe	B1
	disadvantage: half-life shorter so will not provide power for as long	B1